Comparative Evaluation of the SystemDS PowerTransformer

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Abstract

This report evaluates the Apache SystemDS PowerTransformer against no scaling, standard scaling, and robust scaling on regression, classification, and clustering tasks. The implementation supports Yeo–Johnson and Box–Cox transformations, estimates one parameter per feature, standardizes transformed features, and reuses fitted parameters on unseen data. Nine public tabular datasets are evaluated with five fixed random seeds. Distance-based downstream estimators are held constant to isolate the effect of preprocessing. A power transformation ranks first or ties for first on four of the nine datasets, with the strongest gain on the highly skewed Wholesale Customers clustering task. The results also show that no preprocessing method dominates every dataset.

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1 Scope

The experiment addresses two questions. First, does a fitted power transformation improve distance-based prediction or clustering compared with common scaling baselines? Second, what preprocessing cost accompanies that change? The evaluation is comparative rather than a test of universal superiority. All methods use the same data splits, random seeds, and downstream estimator parameters.

2 Implementation under test

SystemDS performs every fitted transformation. The training operation returns the transformed matrix and the fitted parameters:

```
[Y, lambdas, means, scales] = powerTransform(
    X, method="yeo-johnson", standardize=TRUE)
```

The apply operation reuses those parameters without fitting on test data:

```
Ytest = powerTransformApply(
    Xtest, lambdas, means, scales, method="yeo-johnson")
```

Yeo–Johnson is the default because it is defined over the real line. Box–Cox is available through `method="box-cox"` and requires strictly positive features. With standardization enabled, the fitted means and scales are derived from the transformed training matrix and reused by the apply operation.

For a positive value x, the Box–Cox family is

$$T_\lambda(x) = \begin{cases} (x^\lambda - 1)/\lambda, & \lambda \neq 0, \\ \log(x), & \lambda = 0. \end{cases} \tag{1}$$

The Yeo–Johnson family extends the same principle to zero and negative values. SystemDS estimates each feature's parameter by maximizing the profile log-likelihood with Brent optimization.

3 Experimental methodology

3.1 Preprocessing methods

The comparison contains five methods: no scaling, standard scaling, robust scaling based on the median and interquartile range, standardized Yeo–Johnson, and standardized Box–Cox. No offset is added to make Box–Cox applicable. Two Abalone records with zero height are excluded according to the frozen data preparation rule; all other datasets retain their full row set.

3.2 Datasets

The matrix contains two compact core datasets and one preselected skew-focused dataset for each task. The skew-focused configuration was frozen before the downstream results were inspected. Its selection requirements are public tabular data, numeric features without missing values, median absolute feature skewness above 2.0, and at least 5,000 rows for supervised tasks.

Dataset	Task	Rows	Features	Study set	Box-Cox
California Housing	Regression	20,640	8	Core	No
Abalone	Regression	4,175	7	Core	Yes
Parkinsons Telemonitoring	Regression	5,875	19	Skew-focused	No
Breast Cancer Wisconsin	Classification	569	30	Core	No
Wine	Classification	178	13	Core	Yes
HTRU2	Classification	17,898	8	Skew-focused	No
Iris	Clustering	150	4	Core	Yes
Seeds	Clustering	210	7	Core	Yes
Wholesale Customers	Clustering	440	6	Skew-focused	Yes

Table 1: Dataset matrix. Box-Cox is used only for strictly positive feature matrices.

3.3 Downstream estimators

Regression uses `KNeighborsRegressor`; classification uses `KNeighborsClassifier`. Both use five neighbors, uniform weights, Euclidean distance, and one execution thread. Clustering uses K-means with the known class count, 20 initializations, and the experiment seed as `random_state`. These models are sensitive to feature geometry, which makes them suitable for comparing preprocessing methods. No estimator or method-specific hyperparameter tuning is performed.

3.4 Splits, seeds, and leakage control

The fixed seeds are 13, 37, 73, 101, and 149. Regression and classification use 80/20 train/test splits. Classification splits are stratified. Parkinsons Telemonitoring uses a group shuffle split so that recordings from the same subject cannot occur in both training and test data. Each transformer is fitted only on the training features and applied to the corresponding test features. Clustering uses the full dataset for each seed.

3.5 Metrics

Regression is ranked by root mean squared error (RMSE), with R^2 as a secondary metric. Classification is ranked by macro-averaged F1, with accuracy as a secondary metric. Clustering is ranked by adjusted Rand index (ARI), with silhouette score as a secondary metric. RMSE is minimized; the other metrics are maximized. Tables report the mean and sample standard deviation over five runs.

4 Results

4.1 Regression

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
California Housing	1.0607 ± 0.0151	0.6486 ± 0.0160	0.6460 ± 0.0169	0.6114 ± 0.0143	–
Abalone	2.2966 ± 0.0894	2.3575 ± 0.1174	2.3673 ± 0.1183	2.3547 ± 0.1255	2.3580 ± 0.1179
Parkinsons Telemonitoring	12.8276 ± 2.3027	13.0932 ± 2.2760	12.6718 ± 2.1206	12.7222 ± 2.1635	–

Table 2: Regression RMSE results across five fixed seeds (mean ± standard deviation).

Yeo–Johnson obtains the lowest mean RMSE on California Housing: 0.6114 versus 0.6486 for standard scaling, a 5.74% reduction. No scaling leads on Abalone, while robust scaling leads on Parkinsons Telemonitoring. The result is therefore dataset-dependent even within one task and estimator.

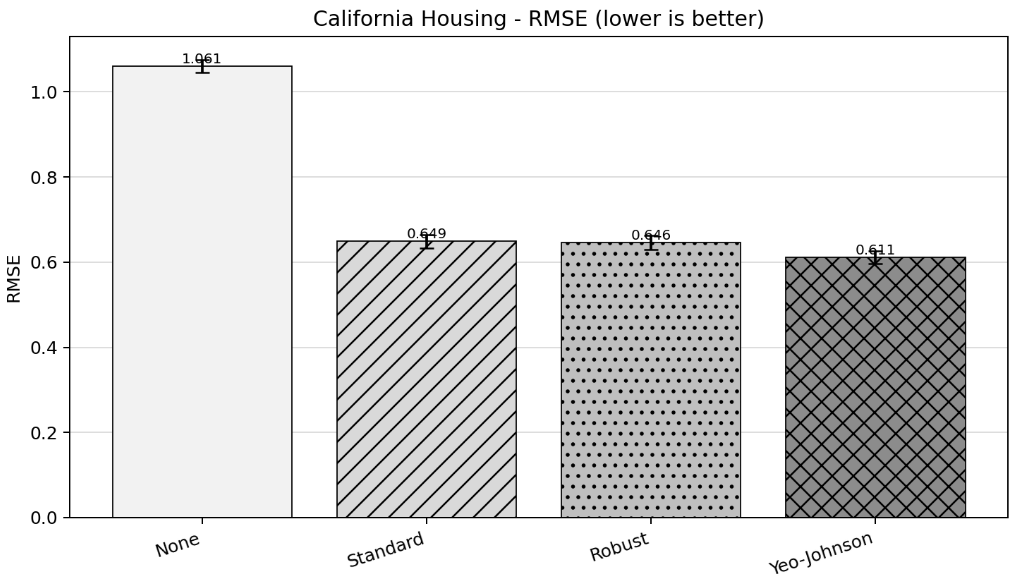

Figure 1: California Housing regression performance. Error bars show one sample standard deviation.

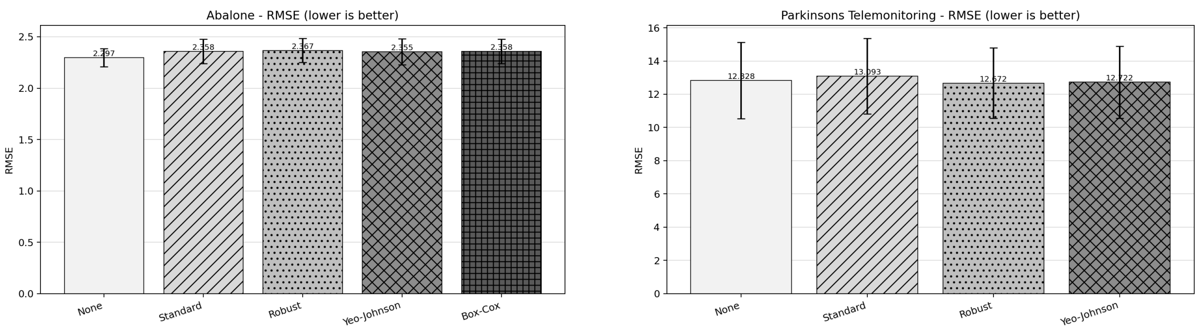

Figure 2: Abalone and Parkinsons Telemonitoring regression performance.

4.2 Classification

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
Breast Cancer Wisconsin	0.9303 ± 0.0271	0.9559 ± 0.0201	0.9560 ± 0.0199	0.9618 ± 0.0136	–
Wine	0.6448 ± 0.0755	0.9734 ± 0.0333	0.9629 ± 0.0452	0.9683 ± 0.0350	0.9683 ± 0.0350
HTRU2	0.9116 ± 0.0087	0.9312 ± 0.0071	0.9339 ± 0.0064	0.9327 ± 0.0042	–

Table 3: Classification MACRO_F1 results across five fixed seeds (mean ± standard deviation).

Yeo–Johnson leads on Breast Cancer Wisconsin with a mean macro-F1 of 0.9618. Standard scaling leads on Wine, and robust scaling leads on HTRU2. The HTRU2 result is close: robust scaling scores 0.9339 and Yeo–Johnson scores 0.9327.

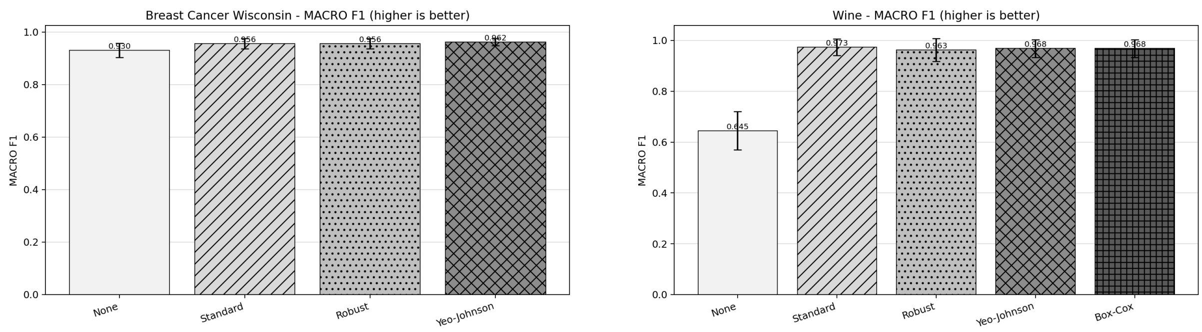

Figure 3: Breast Cancer Wisconsin and Wine classification performance.

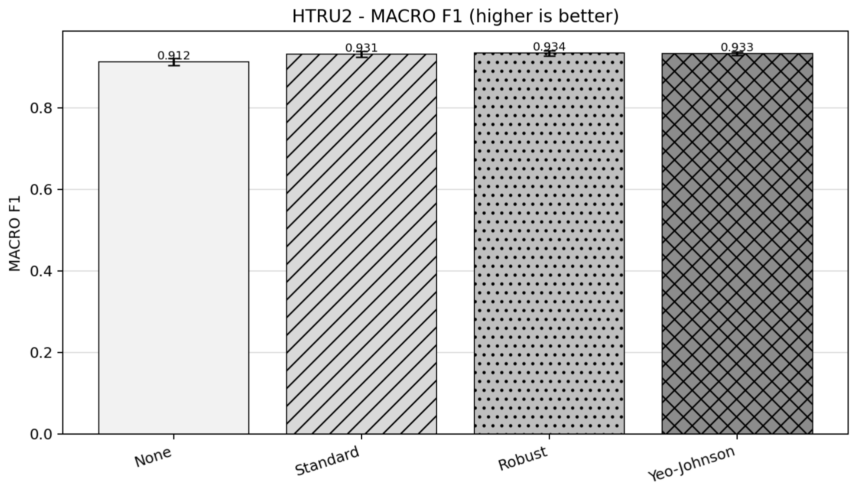

Figure 4: HTRU2 classification performance.

4.3 Clustering

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
Iris	0.7302 ± 0.0000	0.6201 ± 0.0000	0.5803 ± 0.0000	0.6410 ± 0.0000	0.6180 ± 0.0044
Seeds	0.7166 ± 0.0000	0.7733 ± 0.0000	0.7722 ± 0.0000	0.7759 ± 0.0000	0.7759 ± 0.0000
Wholesale Customers	-0.0306 ± 0.0000	0.1873 ± 0.0091	-0.0024 ± 0.0000	0.5264 ± 0.0000	0.5330 ± 0.0000

Table 4: Clustering ARI results across five fixed seeds (mean ± standard deviation).

No scaling leads on Iris. Yeo–Johnson and Box–Cox tie on Seeds at the reported precision. On Wholesale Customers, Box–Cox reaches an ARI of 0.5330 and Yeo–Johnson reaches 0.5264, compared with 0.1873 for standard scaling. This is the clearest case in which reducing extreme right skew changes the K-means geometry in a useful way.

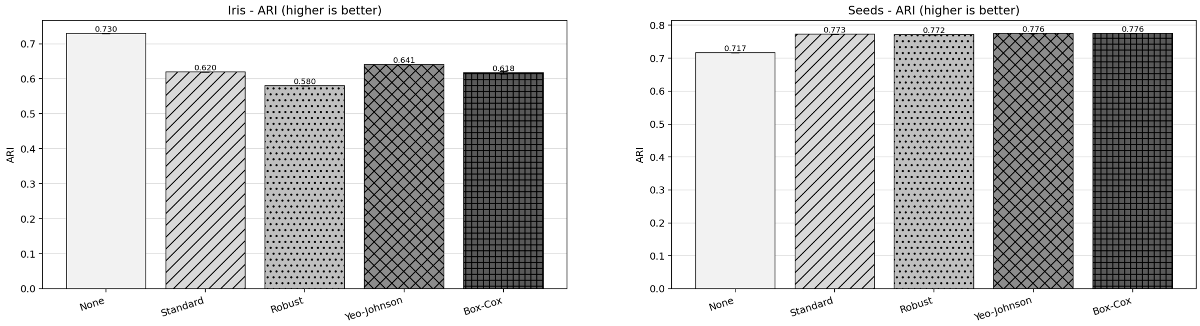

Figure 5: Iris and Seeds clustering performance.

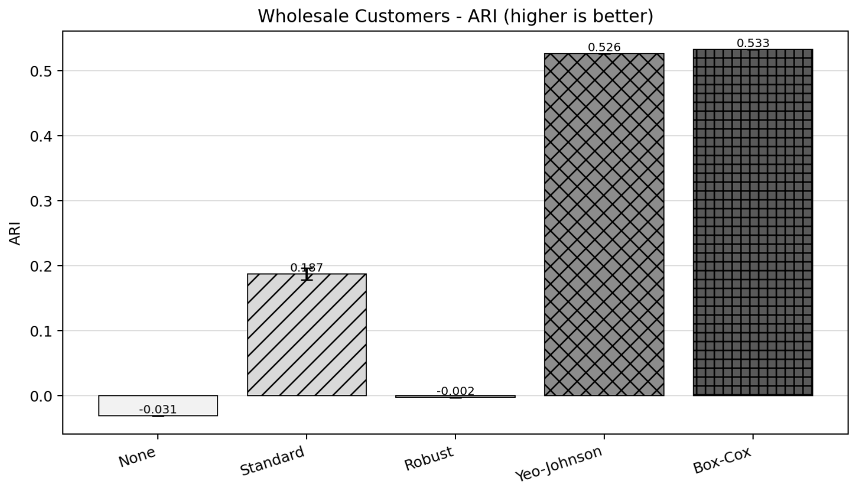

Figure 6: Wholesale Customers clustering performance.

4.4 Preprocessing time

Dataset	Standard	Robust	Yeo-Johnson	Box-Cox
California Housing	0.1430	0.2192	0.8947	–
Abalone	0.0876	0.1536	0.6933	0.3558
Parkinsons Telemonitoring	0.1307	0.2396	1.3546	–
Breast Cancer Wisconsin	0.0867	0.1583	1.3668	–
Wine	0.0740	0.1294	0.8651	0.4845
HTRU2	0.1465	0.2324	1.0706	–
Iris	0.0652	0.1191	0.4112	0.2674
Seeds	0.0696	0.1224	0.6407	0.2788
Wholesale Customers	0.0732	0.1313	0.6394	0.3255

Table 5: Mean SystemDS preprocessing pipeline time in seconds.

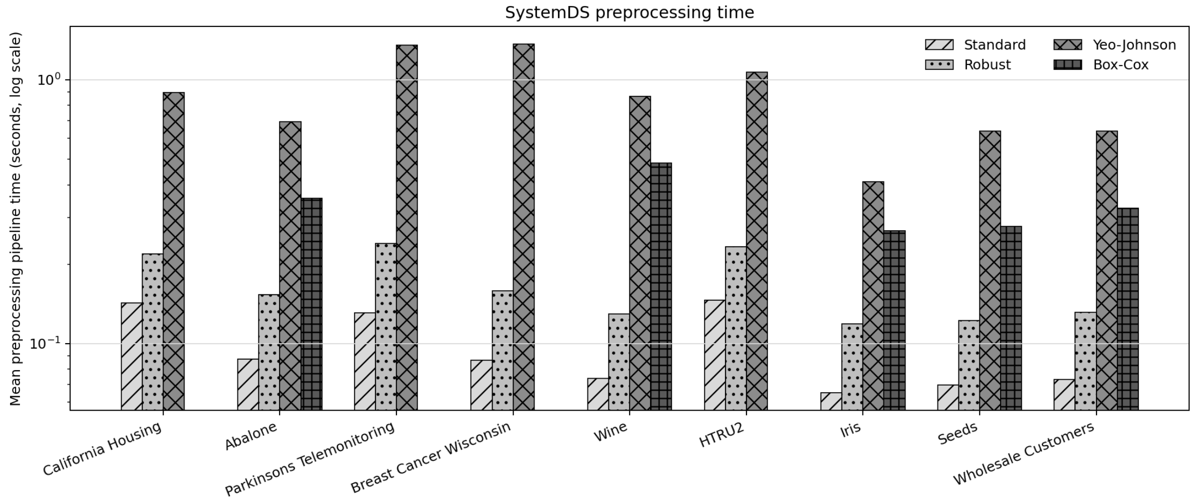

Figure 7: Mean SystemDS preprocessing pipeline time on a logarithmic scale.

Power transformations are slower than standard and robust scaling because every feature requires repeated likelihood evaluations during parameter estimation. The recorded time covers fit, apply, and transformed-matrix writes; it is a pipeline measurement rather than an isolated optimizer microbenchmark.

5 Secondary metrics

The secondary metrics are reported to prevent conclusions from depending on a single score. Internal and external clustering criteria can disagree: silhouette measures Euclidean separation, whereas ARI compares assignments with reference labels.

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
California Housing	0.1554 ± 0.0127	0.6842 ± 0.0093	0.6867 ± 0.0119	0.7194 ± 0.0081	–
Abalone	0.4802 ± 0.0171	0.4524 ± 0.0271	0.4479 ± 0.0259	0.4540 ± 0.0224	0.4523 ± 0.0221
Parkinsons Telemonitoring	-1.1756 ± 1.1695	-1.2943 ± 1.3622	-1.1736 ± 1.3587	-1.1738 ± 1.3065	–

Table 6: Regression R2 results across five fixed seeds (mean ± standard deviation).

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
Breast Cancer Wisconsin	0.9368 ± 0.0227	0.9596 ± 0.0182	0.9596 ± 0.0182	0.9649 ± 0.0124	–
Wine	0.6611 ± 0.0719	0.9722 ± 0.0340	0.9611 ± 0.0465	0.9667 ± 0.0362	0.9667 ± 0.0362
HTRU2	0.9725 ± 0.0026	0.9785 ± 0.0021	0.9792 ± 0.0020	0.9787 ± 0.0013	–

Table 7: Classification ACCURACY results across five fixed seeds (mean ± standard deviation).

Dataset	No scaling	Standard	Robust	Yeo-Johnson	Box-Cox
Iris	0.5528 ± 0.0000	0.4599 ± 0.0000	0.4436 ± 0.0000	0.4572 ± 0.0000	0.4608 ± 0.0003
Seeds	0.4719 ± 0.0000	0.4007 ± 0.0000	0.3913 ± 0.0000	0.4029 ± 0.0000	0.4028 ± 0.0000
Wholesale Customers	0.5115 ± 0.0000	0.5931 ± 0.0057	0.9031 ± 0.0000	0.2950 ± 0.0000	0.2954 ± 0.0000

Table 8: Clustering SILHOUETTE results across five fixed seeds (mean ± standard deviation).

6 Discussion

A power transformation ranks first or ties for first on four datasets: California Housing, Breast Cancer Wisconsin, Seeds, and Wholesale Customers. The benefit is largest when nonlinear marginal skew materially distorts the distance geometry, as in Wholesale Customers. Mildly skewed or already well-separated datasets do not necessarily improve. This is expected: a power transformation changes marginal feature distributions but does not optimize the downstream task directly.

Yeo–Johnson remains the general default because it accepts positive, zero, and negative values. Box–Cox is a useful explicit option for strictly positive features and leads on Wholesale Customers. The results support providing both methods while avoiding a claim that either method should win on every dataset.

7 Limitations

The evaluation uses three datasets per task, five seeds, and one fixed estimator family per task. It is not a universal benchmark. The skew-focused extension contains one preselected dataset per task. No statistical significance test, cross-validation study, missing-value study, inverse transformation, or method-specific hyperparameter search is included. Runtime measurements come from one Apple Silicon machine and small tabular datasets. Transformations run in SystemDS; downstream estimators and metrics run in scikit-learn.